

DisCo-CFL: Calibrated, Fair and Explainable Clustered Federated Learning via Disattenuated Class-Conditional Gradient Signatures

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Abstract

Clustered federated learning (CFL) trains one model per group of clients that share a data distribution. A recent survey of the field [Ben Ali et al., 2026] identifies several open problems: server-side similarities are computed on high-dimensional, round-dependent model updates; clients tend to be grouped by label distribution rather than by concept; the number of clusters is not determined in a principled way; quantity skew breaks existing methods; and the fairness, accountability and transparency of the clustering decision are almost never studied. We propose *DisCo-CFL* (Disattenuated Class-conditional CFL). At a frozen reference model, every client uploads once a count-sketches mean gradient for each class it holds, computed on two random halves of that class. From the two halves the server estimates the reliability of every class signature and applies Spearman’s correction for attenuation, which yields a *calibrated* similarity: it is invariant to label skew and quantity skew by construction, and consistent for the population cosine between class-conditional gradients even under differential-privacy noise. Clusters are formed with a class-wise bottleneck linkage and a single threshold on this calibrated scale, so the number of clusters is found automatically. We prove (i) exact invariance to target and quantity skew, (ii) a finite-sample concentration bound for the calibrated similarity, (iii) safety (conflicting clients are never merged), maximality and exact recovery with a sample complexity that does not depend on cluster size, which is a formal minority-fairness guarantee, (iv) a minimax lower bound that matches the upper bound up to a factor $1/\Delta$ and logarithmic terms, (v) an (ϵ, δ) differential-privacy guarantee with an explicit privacy–utility cost, and (vi) fidelity of the class-level explanations produced by the server. Beyond accuracy, DisCo-CFL outputs per-client assignment certificates with confidence intervals, an audit log of every merge decision, flags for low-evidence and anomalous clients, and an automatic diagnosis of the type of heterogeneity. Experiments on Fashion-MNIST, MNIST and CIFAR-10 cover concept shift on features and targets, label skew, quantity skew, minority groups, newcomers and privacy noise. They compare DisCo-CFL with FedAvg, local training, IFCA, CFL, FL+HC and PACFL and support the theory.

1 Introduction

Federated learning (FL) trains a model across clients that keep their data local [McMahan et al., 2017, Kairouz et al., 2021]. When client data are not identically distributed, a single global model can be a poor fit for everyone [Zhao et al., 2018]. *Clustered* FL relaxes the single-model objective: clients are partitioned into groups with homogeneous distributions and each group trains its own model [Sattler et al., 2021, Ghosh et al., 2020, Briggs et al., 2020]. A recent survey [Ben Ali et al., 2026] organises the field into server-side, client-side and metadata-based methods. It shows that

each family trades privacy, computation, communication and the stability of the clustering metric differently, and it lists open challenges. Our work is driven by five of them.

1. **What to cluster on.** Server-side methods compare full, high-dimensional model updates whose geometry changes from round to round. Several studies report that CFL methods tend to group clients by *label* similarity [Guo et al., 2024], which fragments the data even when clients share the same concept.
2. **Number of clusters.** Every method has a hyperparameter that controls K directly or indirectly, and “fully automatic and stable cluster determination in CFL remains an open research challenge” [Ben Ali et al., 2026].
3. **Quantity skew** is almost never evaluated, even though existing CFL algorithms are known to struggle with it [Ben Ali et al., 2025].
4. **Privacy** is rarely built into the clustering step. Metadata-based methods improve efficiency by sharing statistics that carry a moderate to high privacy risk.
5. **Evaluation is performance-only.** Methods are compared on accuracy. The *quality* of the clustering decision is rarely scrutinised, and neither is its fairness towards small groups or its explainability to the participants.

Our idea. The key observation is simple. At a fixed reference model w , the expected gradient of the loss restricted to class c depends on a client only through its *class-conditional* distribution $\mathcal{P}_i(X \mid Y = c)$. It is therefore unaffected by the client’s label proportions and dataset size. Both kinds of concept shift change it: a feature shift (e.g. rotated inputs) changes $\mathcal{P}(X \mid Y = c)$ for every class, and a target shift (e.g. swapped labels) changes it for the affected classes. Estimated class-conditional gradients, however, are noisy, and noise *attenuates* cosine similarities: two clients with the same concept but few samples look dissimilar. We borrow a century-old idea from psychometrics, Spearman’s correction for attenuation [Spearman, 1904, 1910, Brown, 1910]. Every client splits the samples of each class into two halves, and the agreement between the halves measures the reliability of the signature. Dividing the cross-client cosine by the geometric mean of the two reliabilities gives an estimator of the *population* similarity, on a scale where 1 means “same concept” for every client regardless of how much data it has or how much privacy noise was added.

Contributions.

- **Algorithm.** DisCo-CFL, a one-shot server-side CFL algorithm based on sketched class-conditional gradient signatures, split-half disattenuation, and threshold clustering with a class-wise *bottle-neck* linkage and evidence-ordered merging. K is determined by one threshold on a calibrated scale (Section 4).
- **Theory** (Section 5): exact invariance; a concentration bound; safety, maximality and exact recovery with a size-independent sample complexity; a minimax lower bound; differential privacy with an explicit utility cost; and explanation fidelity.
- **Properties beyond accuracy** (Section 6), which we organise along fairness, accountability, transparency and privacy: minority-group protection, per-client assignment certificates, an audit log, anomaly flags, class-level explanations and an automatic heterogeneity diagnosis.
- **Evaluation** (Section 7) under concept shift on features and targets, label skew, quantity skew, imbalanced groups, newcomers and DP noise, with clustering-quality, fairness, explainability and certificate-calibration metrics in addition to accuracy.

2 Related work

Clustered FL. Server-side methods cluster model updates, e.g. CFL/MTCFL (recursive cosine bipartitioning, [Sattler et al., 2021](#)), FL+HC (one-shot hierarchical clustering after warm-up, [Briggs et al., 2020](#)) and FeSEM [[Long et al., 2023](#)]. Client-side methods let each client pick the model with the lowest loss, e.g. IFCA [[Ghosh et al., 2020](#)], or weight several models, e.g. FedEM [[Marfoq et al., 2021](#)] and FedRC [[Guo et al., 2024](#)]. Metadata-based methods share data statistics; PACFL [[Vahidian et al., 2023](#)], for instance, shares principal subspaces of the raw data. See [Ben Ali et al. \[2026\]](#) for a complete taxonomy. Two recent works also compare clients class by class: FedDAG [[Pramanik et al., 2026](#)] combines gradient similarity with class-wise principal angles between data subspaces, and FedCCFA [[Chen et al., 2024](#)] clusters class-specific classifier vectors under concept drift. DisCo-CFL differs in three ways. (a) It uses neither data subspaces nor classifier weights: it uses sketched class-conditional expected gradients at a frozen reference model. (b) It corrects the noise-induced attenuation of similarities, which is what makes the scale calibrated across sample sizes and privacy levels. (c) It comes with recovery guarantees, a matching lower bound, and accountability and transparency outputs.

Fairness, accountability and transparency in FL. Fairness in FL usually concerns the distribution of accuracy across clients [[Li et al., 2020](#), [Mohri et al., 2019](#)]. In CFL a more basic unfairness arises before training: a small group can be absorbed into a larger one, or a small client can be isolated because its statistics are noisy. We address both at the level of the clustering decision, with a guarantee. Transparency and accountability of *who is grouped with whom, and why* [[Doshi-Velez and Kim, 2017](#), [Barocas et al., 2023](#)] have, to our knowledge, not been treated in CFL.

3 Problem setting

There are N clients. Client i holds $D_i = \{(x_s, y_s)\}_{s=1}^{n_i}$ with labels in $[C]$ drawn from $\mathcal{P}_i(X, Y) = \mathcal{P}_i(Y) \mathcal{P}_i(X | Y)$. Let $S_i = \{c : n_i^c \geq m\}$ be the classes the client holds with at least m samples.

Definition 1 (Concept equivalence and conflicts). Clients i and j *conflict on class* $c \in S_i \cap S_j$ if $\mathcal{P}_i(X | Y = c) \neq \mathcal{P}_j(X | Y = c)$. They are *compatible* if they conflict on no shared class. Clients belong to K latent concept groups $G_1, \dots, G_K$: $\mathcal{P}_i(X | Y = c) = \mathcal{P}^{(k)}(X | Y = c)$ for $i \in G_k$.

The target of clustering is concept equivalence, not equality of joint distributions. If $\mathcal{P}_i(X | Y) = \mathcal{P}_j(X | Y)$, the Bayes classifiers of i and j differ only through the priors $\mathcal{P}_i(Y), \mathcal{P}_j(Y)$, which a shared model can absorb by prior (logit) correction [[Lipton et al., 2018](#), [Menon et al., 2021](#)]. Splitting such clients only reduces the data available to each model. Target skew and quantity skew therefore should *not* create clusters, while concept shift on features and on targets should. In the survey’s taxonomy [[Ben Ali et al., 2026](#)], a rotation of the inputs changes $\mathcal{P}(X | Y = c)$ for all classes, and a label swap $a \leftrightarrow b$ changes it for classes a and b .

4 DisCo-CFL

4.1 Class-conditional gradient signatures (client side)

After T_0 warm-up rounds of FedAvg the server broadcasts a reference model w and the seed of a count-sketch $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^p$ [[Charikar et al., 2002](#), [Weinberger et al., 2009](#)] ($p = 1024$). For a sample

(x, c) let $g(x, c) = \Phi \nabla_w \ell(w; x, c) \in \mathbb{R}^p$. For every class $c \in S_i$ the client randomly splits its class- c samples into halves A_i^c, B_i^c of size h_i^c and sends

$$a_i^c = \frac{1}{h_i^c} \sum_{s \in A_i^c} g(x_s, c), \quad b_i^c = \frac{1}{h_i^c} \sum_{s \in B_i^c} g(x_s, c).$$

This happens once, at a cost of $2p|S_i|$ floats; for our CNN that is about 12% of one model upload. For differential privacy (Section ??), per-sample sketches are clipped to norm C_{clip} and Gaussian noise is added to each half-mean.

4.2 Disattenuated similarity (server side)

Let $m_i^c = (a_i^c + b_i^c)/2$. The split-half reliability and its Spearman–Brown extension to the full mean are

$$r_i^c = \cos(a_i^c, b_i^c), \quad R_i^c = \frac{2r_i^c}{1 + r_i^c} \quad (R_i^c := 0 \text{ if } r_i^c \leq 0). \quad (1)$$

The calibrated similarity of i and j on class c is

$$\hat{\rho}_{ij}^c = \text{clip}_{[-1,1]} \left(\frac{\cos(m_i^c, m_j^c)}{\sqrt{R_i^c R_j^c}} \right), \quad W_{ij}^c = R_i^c R_j^c, \quad (2)$$

where W_{ij}^c is the evidence weight. In the population version of (2) the attenuation factors cancel exactly (Lemma 1), so $\hat{\rho}$ targets $\rho_{ij}^c = \cos(\mu_i^c, \mu_j^c)$, the cosine between expected class-conditional gradients, whatever the sample sizes and noise levels.

4.3 Threshold clustering with bottleneck linkage

For two sets of clients A, B and a class c , let $\text{num}_{AB}^c = \sum_{i \in A, j \in B} W_{ij}^c \hat{\rho}_{ij}^c$ and $\text{den}_{AB}^c = \sum W_{ij}^c$. The linkage is

$$L(A, B) = \min_c \frac{\text{num}_{AB}^c + \lambda}{\text{den}_{AB}^c + \lambda} \quad \text{if } \sum_c \text{den}_{AB}^c \geq e_{\min}, \text{ else } -\infty. \quad (3)$$

The *minimum* over classes encodes the definition of concept equivalence: two groups share a concept only if every class-conditional gradient agrees, so a label swap affecting two of ten classes is not averaged away. The shrinkage λ pulls classes with little evidence towards 1 (“no evidence of a difference”). Starting from singletons, the server repeatedly merges, among the pairs with $L \geq \tau$, the pair with the *largest shared evidence*. Well-supported groups therefore form first, and clients that carry little information on the discriminative classes join them afterwards instead of bridging different groups. Merging stops when no pair reaches τ . Because the scale is calibrated, τ has a fixed meaning: we use $\tau = 0.8$ in *all* experiments, with $\lambda = 0.5$. Clients with total reliability $\sum_c R_i^c < 0.3$ are *low-evidence*: they are excluded from shaping clusters and attached to their most similar cluster afterwards. A singleton cluster is attached to its closest cluster unless the evidence *significantly* separates them (upper confidence bound below τ), in which case it is kept and flagged as *anomalous*. Each cluster then runs FedAvg from w . Algorithm 1 summarises the procedure.

4.4 Transparency and accountability outputs

- **Class-level explanations.** For every pair of clusters the server reports the pooled per-class similarity and the set of *divergent classes* $\hat{D}_{AB} = \{c : \rho_{AB}^c < \tau\}$.

Algorithm 1 DisCo-CFL

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1: Warm-up:  $T_0$  rounds of FedAvg on all clients  $\rightarrow$  reference model  $w$ ; broadcast  $w$  and sketch seed
2: for each client  $i$  in parallel do ▷ once
3:   for  $c \in S_i$  do split class- $c$  samples into halves, send  $(a_i^c, b_i^c)$  (clipped and noised if DP)
4:   end for
5: end for
6: Server:  $R_i^c$  by (1);  $\hat{\rho}_{ij}^c, W_{ij}^c$  by (2)
7: Clusters  $\leftarrow$  singletons of clients with enough evidence
8: while  $\exists A, B$  with  $L(A, B) \geq \tau$  do merge the eligible pair with the largest  $\sum_c \text{den}_{AB}^c$ ; log it
9: end while
10: Attach low-evidence clients and non-significant singletons; flag anomalies
11: Output clusters, certificates, explanations, audit log; run FedAvg within each cluster from  $w$ 

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- **Heterogeneity diagnosis.** If $\hat{D}_{AB}$ contains every shared class, the server reports a global (feature-level) concept shift; if it is a strict subset, a class-specific shift on the listed classes; if $\hat{K} = 1$, it reports that the heterogeneity is explained by target and quantity skew only.
- **Assignment certificates.** Each client receives its bottleneck similarity to its own cluster and to the closest other cluster, the bottleneck classes, and standard errors from the spread over member pairs. A certificate is *confident* if $s_{\text{own}} - z \text{ se} \geq \tau$ and $s_{\text{other}} + z \text{ se} < \tau$ ($z = 2$); otherwise it is *uncertain* or *low-evidence*.
- **Audit log** of every merge (members, linkage, bottleneck class, evidence) and of every attachment or anomaly decision, so that each grouping can be traced and contested.
- **Newcomers** compute their signature at the same frozen w and join the cluster with the highest bottleneck similarity if it reaches τ . Otherwise they are reported as a *novel concept*.

5 Theory

Proofs are in Appendix A. Write $\mu_i^c = \mathbb{E}_{x \sim \mathcal{P}_i(X|Y=c)}[g(x, c)]$ and $a_i^c = \|\mu_i^c\|^2$.

Proposition 1 (Invariance). μ_i^c is a functional of $\mathcal{P}_i(X | Y = c)$ only. Consequently, for compatible clients $\rho_{ij}^c = 1$ on every shared class, whatever their label proportions $\mathcal{P}_i(Y), \mathcal{P}_j(Y)$ and sample sizes n_i, n_j .

Assumption 1 (Noise). Per-sample signatures are $g = \mu_i^c + \Sigma^{1/2}z$ with independent z having independent, zero-mean, unit-variance, K_0 -sub-Gaussian coordinates. Let $\sigma^2 = \text{tr } \Sigma$, and define the effective ranks $r_s = \text{tr } \Sigma / \|\Sigma\|_{\text{op}}$ and $r_e = (\text{tr } \Sigma)^2 / \|\Sigma\|_F^2$.

Assumption 2 (Identifiability). If i, j conflict on c , then $\rho_{ij}^c \leq 1 - \Delta$ for some $\Delta \in (0, 1]$.

For a pair and a class let $h = \min(h_i^c, h_j^c)$, $a = \min(a_i^c, a_j^c)$, and let $\kappa = \sigma^2/(ah)$ be the inverse signal-to-noise ratio of a half-mean.

Lemma 1 (Exact calibration at the population level). Replacing each inner product and squared norm in (1)–(2) by its expectation gives exactly ρ_{ij}^c .

Theorem 1 (Concentration). *Under Assumption 1, with $\kappa \leq 1$, there are constants c_0, C_1 depending only on K_0 such that for $\delta \in (0, 1)$, if $\eta := (\sqrt{\kappa/r_s} + \kappa/\sqrt{r_e})\sqrt{\log(12/\delta)} \leq c_0$, then with probability at least $1 - \delta$*

$$|\hat{\rho}_{ij}^c - \rho_{ij}^c| \leq C_1 \eta, \quad |R_i^c - R_i^{c,*}| \leq C_1 \eta,$$

where $R_i^{c,*} = a_i^c/(a_i^c + \sigma^2/(2h_i^c))$. In contrast, the raw cosine $\cos(m_i^c, m_j^c)$ converges to $\rho_{ij}^c \sqrt{R_i^{c,*} R_j^{c,*}}$, which is biased towards 0 by a factor that depends on the sample sizes.

Theorem 2 (Safety, maximality, recovery). *Let $\gamma = \min(1 - \tau, \tau - 1 + \Delta) > 0$. Suppose all estimates satisfy $|\hat{\rho}_{ij}^c - \rho_{ij}^c| \leq \gamma/2$ (event $\mathcal{E}$), and every conflicting pair on class c has evidence $W_{ij}^c > w_0 := 2\lambda(1 - \tau)/\gamma$. Then the merging phase of Algorithm 1, in any merge order, satisfies:*

- (a) Safety: no cluster ever contains two conflicting clients.
- (b) Maximality: any two output clusters with shared evidence $\geq e_{\min}$ contain a conflicting pair, so no two output clusters can be merged without mixing concepts.
- (c) Exact recovery: if every client holds every class on which some groups differ, and every pair of clients shares evidence $\geq e_{\min}$, the output partition equals $\{G_1, \dots, G_K\}$; in particular $\hat{K} = K$.

Corollary 1 (Sample complexity and minority fairness). *Under Assumptions 1–2, let $\gamma \leq 1/3$ and $\lambda < \gamma/(8(1 - \tau))$. The conclusions of Theorem 2 hold with probability at least $1 - \delta$ as soon as every reported half size satisfies*

$$h \geq h^* := C_2 \frac{\sigma^2}{a} \max \left\{ \frac{\log(12N^2C/\delta)}{r_s \gamma^2}, \frac{\sqrt{\log(12N^2C/\delta)}}{\sqrt{r_e} \gamma}, 1 \right\}. \quad (4)$$

h^* does not depend on the sizes of the groups $|G_k|$, on the number of clients per group, or on the other clients' dataset sizes (apart from $\log N$). A concept group made of a single client is recovered under the same condition as a group of a hundred.

Theorem 3 (Minimax lower bound). *Let $\xi \sim \mathcal{N}(0, \Sigma)$ and $\|\mu\|^2 = a$. For any test that decides from n per-sample signatures of client j , even when μ_i is known exactly, whether $\rho_{ij} = 1$ or $\rho_{ij} = 1 - \Delta$, the worst-case error probability is at least $\frac{1}{4} \exp(-n a \Delta r_s / \sigma^2)$. Error at most δ therefore requires $n \geq \frac{\sigma^2}{a r_s \Delta} \log \frac{1}{4\delta}$.*

With τ in the middle of the admissible interval ($\gamma = \Delta/2$), the leading term of (4) is $O(\sigma^2 \log(N^2C/\delta)/(a r_s \Delta^2))$. The upper and lower bounds therefore agree in their dependence on noise σ^2 , signal a and effective rank r_s , and differ by a factor $1/\Delta$ and a $\log N$ term.

Proposition 2 (Differential privacy and its cost). *Suppose each released half-mean is $\frac{1}{h} \sum_s \text{clip}_{C_{\text{clip}}}(g_s) + \mathcal{N}(0, (2\sigma_{\text{dp}}C_{\text{clip}}/h)^2 I_p)$. (i) Every sample enters exactly one released vector, whose replace-one sensitivity is $2C_{\text{clip}}/h$. The signature is therefore (ε, δ) -DP with respect to replacing one record by another of the same label, with (ε, δ) given by the analytic Gaussian mechanism [Balle and Wang, 2018] at noise multiplier σ_{dp} (parallel composition). The class support S_i is released in the clear. (ii) The DP noise is independent of sampling noise. Lemma 1 therefore still holds, and Theorem 1 applies with the half-mean covariance $\Sigma/h + (2\sigma_{\text{dp}}C_{\text{clip}}/h)^2 I_p$. The recovery guarantee then requires, in addition to (4), $h \gtrsim \sigma_{\text{dp}}C_{\text{clip}}(\sqrt{\log(N^2C/\delta)}/(\sqrt{a}\gamma) + p^{1/4} \log^{1/4}(N^2C/\delta)/\sqrt{a\gamma})$, i.e. the per-class sample size grows linearly in $\sigma_{\text{dp}} \propto 1/\varepsilon$. (iii) Without disattenuation, the raw cosine tends to $\rho \sqrt{R_i R_j}$ with $R \approx a/(a + 2p\sigma_{\text{dp}}^2 C_{\text{clip}}^2/h^2)$. Unless $h \gg \sqrt{p} \sigma_{\text{dp}} C_{\text{clip}}/\sqrt{a}$, same-concept clients look unrelated.*

Proposition 3 (Explanation fidelity). *On $\mathcal{E}$, if two output clusters A, B have pooled evidence $\text{den}_{AB}^c > w_0$ on a class c , then $c \in \hat{D}_{AB}$ if and only if $\mathcal{P}_A(X | Y = c) \neq \mathcal{P}_B(X | Y = c)$. The reported divergent classes are exactly the classes on which the two concepts differ.*

Remark 1 (Scope of the guarantees). Theorem 2 covers the merging phase. The attachment of low-evidence clients and of non-significant singletons is a fairness heuristic: it avoids training a client alone because of noise, may violate safety when evidence is weak, and is logged in the audit trail. Corollary 1 asks for a small λ so that a single pair of clients carries enough evidence. Our default $\lambda = 0.5$ is larger. It makes the method more conservative about splitting on thin evidence, and merges quickly happen between clusters whose pooled evidence far exceeds w_0 . Section 7 reports the sensitivity to λ , including values that satisfy the corollary.

6 Fairness, accountability, transparency and privacy

Fairness. DisCo-CFL is fair in the clustering decision along two axes. *Group size:* Corollary 1 makes recovery independent of cluster size. By contrast, objectives of the k -means type (FL+HC, IFCA) favour large clusters, and IFCA’s analysis depends on the smallest cluster fraction [Ghosh et al., 2020]. *Data size:* small clients are not pushed away by attenuation (Theorem 1), and clients that are merely noisy are attached rather than isolated. We measure worst-group accuracy, the accuracy of the worst 10% of clients, Jain’s index [Jain et al., 1984], and minority recall and purity.

Accountability. Certificates, audit log and anomaly flags make every grouping decision traceable and contestable. We measure the *calibration* of certificates: how often a confident certificate belongs to a client that shares no conflict with its cluster.

Transparency. The explanations and diagnosis tell participants *why* they were separated (“you differ from cluster 2 on classes 3 and 5”), with a fidelity guarantee (Proposition 3). We measure precision, recall and exact match of the divergent-class sets, and the accuracy of the diagnosis.

Privacy. Proposition 2 gives record-level DP for the only extra information released. The signatures are low-dimensional sketches ($p \ll d$) and are uploaded once. Section 7 studies the resulting privacy–utility trade-off.

7 Experiments

TBD

8 Discussion and limitations

TBD

9 Conclusion

DisCo-CFL replaces ad hoc similarity heuristics in clustered FL with a *calibrated* statistic: the disattenuated cosine between class-conditional gradients. This single change yields invariance to label and quantity skew, a threshold with a fixed meaning (and hence an automatic K), recovery

guarantees that do not depend on group size, consistency under privacy noise, and explanations whose fidelity can be proven. The algorithm reports assignment certificates, an audit log, anomaly flags and class-level explanations together with the clusters, which extends the evaluation of CFL from accuracy alone to fairness, accountability, transparency and privacy.

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A Proofs

Throughout, $C, C_1, C_2, \dots$ and c_0 denote constants that depend only on the sub-Gaussian constant K_0 and may change from line to line. We fix a class c and drop it from the notation when no confusion arises.

A.1 Proof of Proposition 1

The reference model w , the sketch Φ and the label c are the same for all clients, so $\mu_i^c = \int \Phi \nabla_w \ell(w; x, c) d\mathcal{P}_i(x | Y = c)$ is determined by $\mathcal{P}_i(X | Y = c)$ alone. Neither the proportion $\mathcal{P}_i(Y = c)$ nor the number of samples enters. If i and j are compatible, then $\mathcal{P}_i(X | Y = c) = \mathcal{P}_j(X | Y = c)$ on each shared class, hence $\mu_i^c = \mu_j^c$ and $\rho_{ij}^c = 1$. Per-sample clipping (Section ??) is applied identically by all clients, so the same holds for the expected clipped signature. $\square$

A.2 Proof of Lemma 1

Write the half-means as $a_i = \mu_i + \zeta_{i1}$, $b_i = \mu_i + \zeta_{i2}$, where ζ_{i1}, ζ_{i2} are independent, zero-mean and have covariance Σ_i/h_i (any covariance works, and it may differ across clients). Let $t_i = \text{tr } \Sigma_i/h_i$. Then $\mathbb{E}\langle a_i, b_i \rangle = a_i$ and $\mathbb{E}\|a_i\|^2 = \mathbb{E}\|b_i\|^2 = a_i + t_i$, so the population split-half reliability is $r_i^* = a_i/(a_i + t_i)$ and

$$R_i^* = \frac{2r_i^*}{1 + r_i^*} = \frac{a_i}{a_i + t_i/2}.$$

For the full means, $\mathbb{E}\|m_i\|^2 = a_i + t_i/2 = a_i/R_i^*$ and, by independence across clients, $\mathbb{E}\langle m_i, m_j \rangle = \langle \mu_i, \mu_j \rangle$. The population cosine is therefore $\langle \mu_i, \mu_j \rangle \sqrt{R_i^* R_j^*} / \sqrt{a_i a_j}$. Dividing by $\sqrt{R_i^* R_j^*}$ gives ρ_{ij} exactly. $\square$

A.3 Proof of Theorem 1

Noise representation. A half-mean noise vector is $\zeta = (\Sigma/h)^{1/2} \bar{z}$ with $\bar{z} = h^{-1/2} \sum_s z_s$. Its coordinates are independent, zero-mean, unit-variance and CK_0 -sub-Gaussian.

Statistics. $\hat{\rho}_{ij}$ and R_i are functions of the nine statistics $X_1 = \langle a_i, b_i \rangle$, $X_2 = \|a_i\|^2$, $X_3 = \|b_i\|^2$, $X_4 = \|m_i\|^2$ (and $X_5, \dots, X_8$ for j) and $X_9 = \langle m_i, m_j \rangle$. We normalise them as $x_k = X_k/a_i$ for the statistics of i , $x_k = X_k/a_j$ for those of j , and $x_9 = X_9/\sqrt{a_i a_j}$. Their expectations are $x^* = (1, 1 + \kappa_i, 1 + \kappa_i, 1 + \kappa_i/2, \dots, \rho_{ij})$ with $\kappa_i = \text{tr } \Sigma/(a_i h_i) \leq \kappa$. Each statistic has the form constant + linear term $\langle \mu, \zeta \rangle$ + quadratic or bilinear form in ζ 's.

Linear terms. $\langle \mu_i, \zeta \rangle$ is sub-Gaussian with variance proxy $\leq CK_0^2 \mu_i^\top \Sigma \mu_i / h \leq CK_0^2 a_i \|\Sigma\|_{\text{op}} / h$. After normalisation by a_i (or by $\sqrt{a_i a_j}$ for cross terms), with probability $1 - \delta'$ it is at most $C\sqrt{\|\Sigma\|_{\text{op}}/(ah)}\sqrt{\log(2/\delta')} = C\sqrt{\kappa/r_s}\sqrt{\log(2/\delta')}$, because $\|\Sigma\|_{\text{op}} = \sigma^2/r_s$.

Quadratic and bilinear terms. $\|\zeta\|^2 - \text{tr}(\Sigma/h)$ and $\langle \zeta, \zeta' \rangle$ for independent ζ, ζ' are centred quadratic forms $\bar{Z}^\top M \bar{Z}$ of the concatenated vector $\bar{Z}$, with $\|M\|_F \leq \|\Sigma\|_F/h$ and $\|M\|_{\text{op}} \leq \|\Sigma\|_{\text{op}}/h$. The Hanson–Wright inequality [Rudelson and Vershynin, 2013] gives, with probability $1 - \delta'$, a deviation of at most $CK_0^2(\|M\|_F\sqrt{\log(2/\delta')} + \|M\|_{\text{op}}\log(2/\delta'))$. After normalisation by a this is at most $C(\kappa/\sqrt{r_e})\sqrt{\log(2/\delta')} + C(\kappa/r_s)\log(2/\delta')$, using $\|\Sigma\|_F = \sigma^2/\sqrt{r_e}$. The second term is at most $C\eta^2 \leq Cc_0\eta$.

Union bound. Taking $\delta' = \delta/6$ over the at most six independent noise terms that appear in each group of statistics, we obtain $\|x - x^*\|_\infty \leq C\eta$ with probability $1 - \delta$.

Smoothness. Write $\hat{\rho} = F(x)$ with $r_i = x_1/\sqrt{x_2 x_3}$, $R_i = 2r_i/(1 + r_i)$ (similarly for j), and $\hat{\rho} = x_9/\sqrt{x_4 x_8 R_i R_j}$ before clipping. By Lemma 1, $F(x^*) = \rho_{ij}$. For $\kappa \leq 1$ and $\|x - x^*\|_\infty \leq Cc_0$ with c_0 small enough, the argument stays in a compact set on which $x_2, x_3, x_4 \in [1/2, 3]$, $r_i \geq 1/5$ and $R_i \geq 1/3$. On that set F is continuously differentiable with gradient bounded by an absolute constant. Hence $|F(x) - F(x^*)| \leq C_1\eta$, and the same argument gives the bound for R_i . Clipping to $[-1, 1]$ can only reduce the error, because $\rho_{ij} \in [-1, 1]$. For the raw cosine, $x_9/\sqrt{x_4 x_8} \rightarrow \rho_{ij}/\sqrt{(1/R_i^*)(1/R_j^*)} = \rho_{ij}\sqrt{R_i^* R_j^*}$. $\square$

A.4 Proof of Theorem 2

We use two facts, both on $\mathcal{E}$.

Claim 1 (conflicting clusters cannot be linked). Let A, B be conflict-free and suppose some $i \in A, j \in B$ conflict on class c . Since A is conflict-free, all members of A holding c share one class-conditional $\mathcal{P}_A^c$; likewise for B , with $\mathcal{P}_A^c \neq \mathcal{P}_B^c$. Hence every pair in $A \times B$ holding c conflicts on c . By Assumption 2 and $\mathcal{E}$, each such pair has $\hat{\rho} \leq 1 - \Delta + \gamma/2$. Thus $\text{num}_{AB}^c \leq (1 - \Delta + \gamma/2) \text{den}_{AB}^c$ and

$$\frac{\text{num}_{AB}^c + \lambda}{\text{den}_{AB}^c + \lambda} < \tau \iff \text{den}_{AB}^c(\tau - 1 + \Delta - \gamma/2) > \lambda(1 - \tau).$$

Since $\tau - 1 + \Delta \geq \gamma$, the left side is at least $\text{den}_{AB}^c \gamma/2 \geq W_{ij}^c \gamma/2 > w_0 \gamma/2 = \lambda(1 - \tau)$. The minimum over classes in (3) is therefore below τ .

Claim 2 (compatible clusters are always linkable). If $A \cup B$ is conflict-free, then every pair in $A \times B$ that holds a class c has $\rho^c = 1$ (Proposition 1), so $\hat{\rho}^c \geq 1 - \gamma/2$. Hence every class term of (3) is at least $1 - \gamma/2 \geq 1 - (1 - \tau)/2 = (1 + \tau)/2 > \tau$; classes without evidence contribute 1. So $L(A, B) > \tau$ whenever the shared evidence is at least $e_{\min}$.

(a) Singletons are conflict-free. By Claim 1 a merge is only performed between clusters whose union is conflict-free, so the invariant is preserved by every merge, in any order. (b) At termination every pair of clusters with shared evidence $\geq e_{\min}$ has $L < \tau$; by Claim 2 their union is not conflict-free. (c) Under full support, two clients of different groups $k \neq l$ both hold a class on which $\mathcal{P}^{(k)}$ and $\mathcal{P}^{(l)}$ differ, so they conflict. By (a) each output cluster is therefore contained in a single group. Two output clusters inside the same group have a conflict-free union and share evidence $\geq e_{\min}$, which contradicts (b). Each group thus forms exactly one cluster. $\square$

A.5 Proof of Corollary 1

There are at most $N^2 C/2$ pairwise class estimates and NC reliabilities; we apply Theorem 1 to each with $\delta/(N^2 C)$ and write $\mathsf{L} = \log(12N^2 C/\delta)$. The requirement $C_1\eta \leq \gamma/2$ holds if $\sqrt{\kappa/r_s}\sqrt{\mathsf{L}} \leq$

$\gamma/(4C_1)$ and $(\kappa/\sqrt{r_e})\sqrt{L} \leq \gamma/(4C_1)$. With $\kappa = \sigma^2/(ah)$, these are the two terms of (4); the side conditions $\kappa \leq 1$ and $\eta \leq c_0$ are absorbed into C_2 and the additional requirement $h \geq \sigma^2/a$. For the evidence condition: on the event, $R \geq R^* - \gamma/2 \geq 2/3 - \gamma/2$ (since $\kappa \leq 1$). For $\gamma \leq 1/3$ this gives $W \geq 1/4$, so $W > w_0$ holds whenever $\lambda < \gamma/(8(1 - \tau))$. None of these conditions involves $|G_k|$. $\square$

A.6 Proof of Theorem 3

Let e_1 be a top eigenvector of Σ ($\Sigma e_1 = \|\Sigma\|_{\text{op}} e_1$) and $e_2 \perp e_1$ a unit vector. With $s^2 = \Delta/2$ put $u = \sqrt{a}(\sqrt{1 - s^2} e_2 + s e_1)$ and $v = \sqrt{a}(\sqrt{1 - s^2} e_2 - s e_1)$. Then $\|u\|^2 = \|v\|^2 = a$, $\cos(u, v) = 1 - 2s^2 = 1 - \Delta$ and $u - v = 2\sqrt{a} s e_1$. Under H_0 the n samples of client j are i.i.d. $\mathcal{N}(u, \Sigma)$; under H_1 they are $\mathcal{N}(v, \Sigma)$. Moreover $\mu_i = u$ is known. Then

$$\text{KL}(\mathcal{N}(u, \Sigma)^{\otimes n} \parallel \mathcal{N}(v, \Sigma)^{\otimes n}) = \frac{n}{2}(u - v)^\top \Sigma^{-1}(u - v) = \frac{n}{2} \cdot \frac{4as^2}{\|\Sigma\|_{\text{op}}} = \frac{n a \Delta r_s}{\sigma^2}.$$

By the Bretagnolle–Huber inequality (Tsybakov, 2009, Thm. 2.2), every test ψ satisfies $\max_{H \in \{0,1\}} \Pr_H(\psi \neq H) \geq \frac{1}{4} \exp(-\text{KL})$. A procedure that sees only half-means (as DisCo-CFL does), or that does not know μ_i , has access to less information, so the bound applies to it as well. $\square$

A.7 Proof of Proposition 2

(i) Replacing one record by another of the same label changes one clipped summand, hence the half-mean by at most $2C_{\text{clip}}/h$ in ℓ_2 . The added noise has standard deviation σ_{dp} times this sensitivity, so the analytic Gaussian mechanism [Balle and Wang, 2018] gives (ε, δ) -DP for that vector. Every record enters exactly one released vector, so by parallel composition the whole signature is (ε, δ) -DP. The server’s computations are post-processing [Dwork and Roth, 2014]. The class supports and half sizes are released and are not protected. (ii) The DP noise is zero-mean and independent of everything else. The proof of Lemma 1 used only these two properties, so it still holds with $t_i = \text{tr} \Sigma_{\text{clip}}/h + p(2\sigma_{\text{dp}} C_{\text{clip}}/h)^2$. In the proof of Theorem 1, the Gaussian DP component adds a linear term with normalised standard deviation $2\sigma_{\text{dp}} C_{\text{clip}}/(h\sqrt{a})$ and a quadratic term with normalised Frobenius scale $4\sqrt{p} \sigma_{\text{dp}}^2 C_{\text{clip}}^2/(h^2 a)$. Requiring each to be at most $\gamma/(4C_1 \sqrt{L})$ gives the two stated conditions. (iii) This follows from the last statement of Theorem 1 with the inflated t_i . $\square$

A.8 Proof of Proposition 3

Output clusters are conflict-free (Theorem 2(a)), so the members of A holding c share $\mathcal{P}_A^c$, and likewise for B . If $\mathcal{P}_A^c \neq \mathcal{P}_B^c$, the computation of Claim 1 with $\text{den}_{AB}^c > w_0$ gives a pooled value below τ , so $c \in \hat{D}_{AB}$. Otherwise all pairs have $\rho^c = 1$ and the pooled value is at least $(1 + \tau)/2 > \tau$ (Claim 2), so $c \notin \hat{D}_{AB}$. $\square$